AUTOSAR HLD Document Analysis Assistant

A local Retrieval-Augmented Generation (RAG) system with page-level citations, refusal on missing evidence, and human review

Technical Report	Capstone project, TATA Technologies Pulse induction programme
Case study	Case Study 1: AUTOSAR HLD Document Analysis Assistant
Student	PRADEEPA
University / Register No.	SASTRA / 127156121
Programme	CSE (AI & DS)
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Summary. Architects and engineers spend considerable time searching long AUTOSAR PDFs. This project indexes a public AUTOSAR specification locally and lets an engineer ask questions in natural language. Every answer is grounded in retrieved text, cites the source page, is checked for citation validity, and is refused when the evidence is missing. Each answer is reviewed by an engineer and the decision is stored in an audit log. On a test set of 8 answerable and 2 out-of-scope questions, the expected page was retrieved for 8 of 8 questions, all 8 answers carried valid citations, 6 of 8 answers were judged correct on manual review, and both out-of-scope questions were refused.

1. Problem statement and intended users

AUTOSAR architecture and High-Level Design documents are hundreds of pages of unstructured PDF. Finding how a layer, interface or mechanism is defined means manual reading and cross-referencing, which is slow and error-prone, and new team members take a long time to become productive. General-purpose chatbots are not a safe substitute because they can answer from memory, invent details, and give no verifiable source.

Goal. Build an assistant that answers questions about an AUTOSAR document using only that document, shows where each answer comes from, declines to answer when the document does not contain the answer, and keeps a human in the loop. The project follows Case Study 1 (AUTOSAR HLD Document Analysis Assistant) of the Automotive Engineering AI case study document, using its recommended pilot stack.

Intended users. System architects, AUTOSAR architects, software developers, integration engineers, test engineers and engineering managers. Typical uses are design-review preparation, onboarding, and quickly locating the definition of a layer, interface or mechanism.

2. Input data and knowledge base

The knowledge base is the public AUTOSAR Classic Platform document *Layered Software Architecture* (file AUTOSAR_EXP_LayeredSoftwareArchitecture, release R22-11, 182 pages), downloaded from autosar.org and stored as `Input_Data/autosar_doc.pdf`. A public document was chosen deliberately: it raises no confidentiality concern and it covers the software layers, the Runtime Environment (RTE), basic software modules, mode management and end-to-end (E2E) protection, which gives a wide range of questions.

Processing. The PDF is parsed page by page with PyMuPDF. Each page is split into chunks of 300 words with a 50-word overlap, discarding fragments shorter than 100 characters. Chunks never cross a page

boundary, so every chunk has exactly one page number that can be cited. Each chunk is stored with its page number and source file name. This produced **184 chunks**.

Limits of the input. Only extractable text is indexed. Tables and diagrams are flattened to plain text, and scanned pages would need OCR, which is not implemented.

3. Solution architecture

The system is a Retrieval-Augmented Generation (RAG) pipeline. Documents are embedded once and stored in a local vector database. At question time the most relevant chunks are retrieved and the language model is instructed to answer only from them. Figure 1 shows the end-to-end workflow.

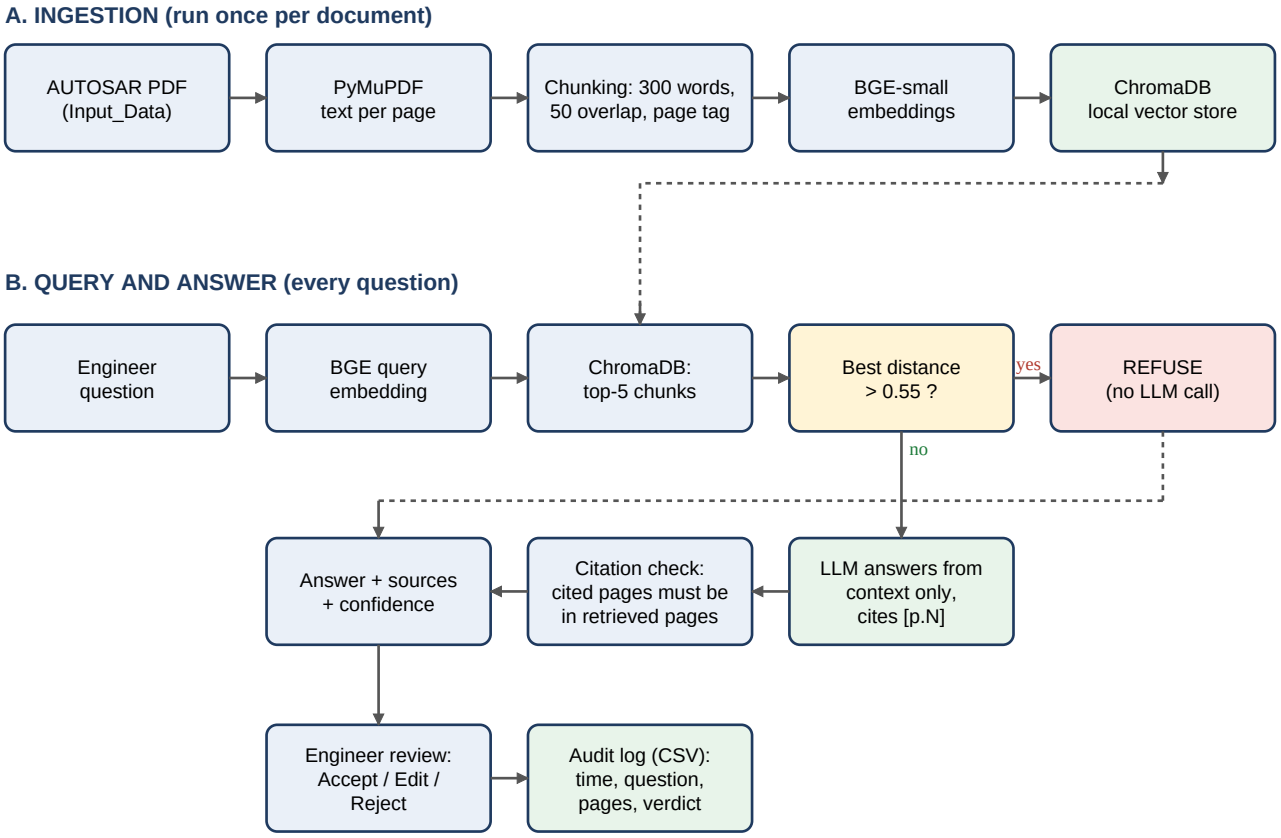

Figure 1. Ingestion and query workflow, including refusal, citation check and human review.

Components.

Layer	Choice	Role
Parsing	PyMuPDF	Extracts text page by page
Embeddings	BAAI/bge-small-en-v1.5	Converts chunks and questions to vectors (BGE query prefix used for questions)
Vector store	ChromaDB (persistent, cosine)	Local semantic retrieval with page metadata
LLM	openai/gpt-oss-20b via Groq API (evaluation); Llama 3.2 3B via Ollama (local option)	Generates the grounded, cited answer; backend chosen by environment variable
Interface	Streamlit	Ask, Component inventory and Audit log tabs
Evaluation	Python script (evaluate.py)	Runs the test set and computes metrics

4. Implementation and configuration

The code is organised so each file has one job.

File	Responsibility
Code/config.py	All tunable settings in one place
Code/ingest.py	PDF to page-aware chunks, embeddings, ChromaDB collection
Code/rag.py	Retrieval, prompts, refusal logic, LLM call, citation validation, confidence, component extraction, audit logging
Code/app.py	Streamlit interface
Code/evaluate.py	Automated evaluation and summary table

Configuration used.

Parameter	Value
Chunk size / overlap	300 words / 50 words
Top-k retrieved chunks	5
Refusal threshold	Best cosine distance > 0.55 means refuse
High-confidence distance	< 0.35
Embedding model	BAAI/bge-small-en-v1.5
LLM temperature	0.1
LLM used for the reported evaluation	openai/gpt-oss-20b (Groq API)

Question-time logic.

- The question is embedded (with the BGE retrieval prefix) and the five nearest chunks are retrieved.
- If the best distance is greater than 0.55 the system answers *I could not find this in the document.* without calling the LLM.
- Otherwise the chunks, each tagged with its page, are placed in the prompt. The system prompt tells the model to use only the context, cite pages as [p.N], treat the context as untrusted data and never as instructions, and reply with the refusal sentence if the answer is absent.
- The application extracts the cited page numbers and checks that each was among the retrieved pages. Confidence is High if citations are valid and the best distance is below 0.35, Medium if citations are valid, and Low otherwise (with a warning to verify manually).
- The engineer records Accept, Edit needed or Reject, and the decision is appended to an audit log.

Backend choice. The LLM backend is selected by environment variables, so the same code can run a local model through Ollama or any OpenAI-compatible hosted API. A local Llama 3.2 3B run worked but answered in about 200 seconds on a CPU-only laptop, which was too slow to evaluate and demonstrate. The hosted gpt-oss-20b model was therefore used for the reported evaluation and the demonstration. Because the source document is public, this does not expose any confidential data. For confidential documents the local option is the intended configuration.

Sample interactions (transcribed from the running application).

Question	Response	Result
What is the role of the RTE?	The AUTOSAR Runtime Environment (RTE) is the layer that supplies communication services to the application software. It acts as the interface between AUTOSAR Software Components (and Sensor/Actuator components) and the underlying services... [p.19]	Confidence High, valid citation, 28.3 s (includes one-time model load)

Question	Response	Result
What does the Complex Drivers layer do?	The Complex Drivers layer provides the possibility to integrate special purpose functionality, such as drivers for devices that are not specified within AUTOSAR, with very high timing constraints or for migration purposes, etc. [p.17]	Confidence High, valid citation
What is the capital of France?	I could not find this in the document.	Refused by distance threshold (0.572), no LLM call

5. Evaluation steps and results

Method. A test set of 10 questions was written over the indexed document: 8 answerable questions, each with an expected page and expected keywords, and 2 out-of-scope questions (*What is the capital of France?* and a chocolate-cake recipe request) to test refusal. `evaluate.py` sends every question through the full pipeline and records the retrieved pages, best distance, refusal status, citation validity, keyword match and latency. Each answerable answer was then checked manually against the PDF.

Answerable questions.

#	Question	Expected keyword(s)
1	What is the role of the RTE?	communication
2	What does the Complex Drivers layer do?	special purpose
3	How does the RTE handle vehicle mode requests and mode switches?	mode
4	How does the E2E protection wrapper work with the RTE?	E2E
5	What are the three main software layers in the AUTOSAR architecture?	Application; Runtime; Basic Software
6	What is the Services Layer in AUTOSAR Basic Software?	service
7	What is the purpose of the Microcontroller Abstraction Layer?	microcontroller
8	What does the ECU Abstraction Layer do?	ECU

Results.

Metric	Result
Retrieval hit rate (expected page in top-5)	8 / 8
Groundedness (answers with valid citations)	8 / 8
Keyword match (automatic proxy)	6 / 8
Manual answer accuracy (checked against the PDF)	6 / 8 (75%)
Correct refusals on out-of-scope questions	2 / 2
Wrongful refusals on answerable questions	0 / 8
Average latency (hosted LLM)	6.4 s

Analysis. Relevant questions retrieved their evidence at distances between 0.11 and 0.34, while the two out-of-scope questions scored 0.572 and 0.536. On this test set the threshold of 0.55 therefore separates relevant from irrelevant queries. The France question was refused by the distance threshold alone, without calling the LLM. The cake question passed the threshold and was refused by the model because of the prompt rule, which shows the two layers of defence working together. The automatic keyword check matched 6 of 8 answers. Manual review against the PDF judged 6 of the 8 answers correct (75%), the same count as the automatic keyword check. The two answers that were not fully correct were not retrieval failures: the expected page was retrieved for all 8 questions and all 8 answers carried valid citations. The shortfall

therefore lies in answer generation, where the response was incorrect or incomplete even though the relevant context had been supplied. This shows that strong retrieval and citation scores do not guarantee a complete answer, which is why human review is part of the design.

Limitations of the evaluation. The test set is small (10 questions). The first four answerable questions were chosen from pages seen earlier in the system's own retrieved results, which may flatter the hit rate. The groundedness check confirms that every cited page was among the retrieved pages. It does not prove that each sentence is supported by the page it cites. For example, one RTE answer carried a single citation [p.19] but included a statement about mode-request distribution that came from another retrieved page.

6. Responsible AI measures

Area	Measure adopted
Grounding	The prompt restricts the model to the retrieved context and requires a page citation after every claim. The application verifies that every cited page was actually retrieved and flags answers that fail this check.
Refusal on missing evidence	If the best retrieved chunk is farther than 0.55, the system replies *I could not find this in the document.* without calling the LLM. The prompt contains the same rule for cases that pass the threshold.
Privacy and security	Parsing, embeddings and the vector database run locally. The source is a public document, so a hosted LLM was acceptable for this prototype. The code also supports a fully local model for confidential documents. Retrieved text is treated as untrusted data in the prompt to reduce prompt-injection risk. API keys are supplied by environment variable and are not stored in project files.
Human oversight	Each answer is shown with its confidence, citation validity and retrieved evidence. The engineer records Accept, Edit needed or Reject, and every decision is stored in an audit log (timestamp, question, answer preview, cited pages, verdict). AI-extracted component inventories are labelled as requiring review against the source pages.
Transparency	Retrieved source passages, page numbers and distances are visible in the interface, so users can verify any answer.

7. Innovation highlights

- **Two-layer refusal:** a distance threshold stops irrelevant questions before the LLM is called, and a prompt rule catches the rest.
- **Citation validator:** answers whose citations do not match the retrieved pages are flagged as low confidence.
- **Confidence indicator** combining retrieval distance with citation validity.
- **Page-bounded chunking** so every chunk and every answer can be traced to a single page.
- **BGE query prefix** to improve retrieval quality for question-style queries.
- **Human-review workflow with a persistent audit log** supporting traceability.
- **Component inventory extractor:** an optional tool that asks the LLM for components, ports and interfaces as structured JSON from the most relevant sections, with a CSV download and a review warning.
- **Pluggable LLM backend:** local Ollama or any OpenAI-compatible API, selected through environment variables, with automatic retry on rate-limit responses.

8. Limitations and future work

- Tables and diagrams are flattened to text and parse poorly; scanned pages are not supported because OCR is not implemented.

- Citation checking works at page level, not sentence level, so a correct citation does not guarantee that every sentence is supported.
- The refusal threshold was tuned on a small test set and may need re-tuning for other documents.
- Small local models are slow on CPU-only hardware and weaker at multi-step reasoning.
- Only one document is indexed at a time and revision comparison is not implemented.

Future work. Table-aware parsing and OCR, sentence-level citation verification, a larger and independently written test set, document revision comparison and inconsistency reporting, a FastAPI service layer with role-based access control, and a graph store for component and interface traceability, as outlined in the reference roadmap.

9. References and AI-tool usage

- AUTOSAR, *Layered Software Architecture*, AUTOSAR Classic Platform (AUTOSAR_EXP_LayeredSoftwareArchitecture), autosar.org.
- TATA Technologies, *Automotive Engineering AI: Standard Project Case Study Document*, Version 1.0, September 2026.
- BAAI, bge-small-en-v1.5 embedding model (Hugging Face).
- ChromaDB; PyMuPDF; Streamlit; Ollama (Llama 3.2); Groq API serving OpenAI gpt-oss-20b.

AI tools used (as declared in the signed AI-tool usage declaration):

AI model / tool	Purpose	Artifact affected	Verified by student
OpenAI gpt-oss-20b (Groq API)	Answer generation	Application runtime output	Answers checked against PDF pages in the evaluation
BAAI/bge-small-en-v1.5	Text embeddings	ingest.py, retrieval	Retrieval hit rate measured on the test set
Claude (Anthropic)	Project planning and code scaffolding	Code/ and documentation	All code run, tested and understood by the student